

INCIDENT REPORT: Network Attack Detection and Analysis

Date: 9 January 2026 | **Analyst:** Paul Umeh Ogonna | **Lab Environment:** Isolated Virtual Network (Kali Linux & Cowrie Honeypot)

Target IP: 192.168.2.129 | **Attack Source:** 192.168.2.130 (Kali Linux VM)

Executive Summary

A simulated network intrusion was conducted against a Cowrie SSH honeypot to demonstrate attack detection capabilities. The attacker performed reconnaissance scanning via Nmap, followed by successful SSH authentication using weak credentials (root/12345). All activity was captured through honeypot logs, Wireshark packet analysis, and Nmap output, demonstrating effective visibility into reconnaissance and intrusion attempts.

Incident Timeline

- 22:30** - Nmap port scan initiated from 192.168.2.130
- 22:33:30** - SSH service discovered on TCP 2222, OpenSSH 9.2p1 identified
- 22:33:31** - Login attempt with root/12345 succeeded (simulated by honeypot)
- 22:36:31** - Session terminated | **Duration:** ~3 minutes

Evidence Analysis

Nmap Scan Results: Service detection revealed open port 2222 running OpenSSH 9.2p1 on Debian. The scan confirmed SSH was discoverable despite using a non-standard port, demonstrating that obscurity alone is insufficient security.

Wireshark Capture: Packet analysis showed TCP handshakes, SSH protocol negotiation, encrypted authentication traffic, and session data. The `cowrie_attack_capture.pcapng` file provides complete forensic evidence of the attack sequence.

Honeypot Logs: Cowrie successfully logged the connection from 192.168.2.130, captured authentication attempts, and recorded session activity. Sample log entry:

```
2026-01-08T22:33:30 [192.168.2.130] login attempt [root/12345] succeeded
2026-01-08T22:33:31 Terminal Size: 76x43, Handling pty request
2026-01-08T22:36:31 avatar root logging out, connection lost
```

Impact Assessment

While conducted in a controlled environment, this attack represents realistic threats including: unauthorized access via weak credentials, potential lateral movement, privilege escalation opportunities, and data exfiltration risks. The honeypot prevented actual system compromise while capturing valuable attacker behavior.

Recommendations

1. **Authentication Hardening:** Implement SSH key-based authentication, disable password login, enforce MFA, and disable direct root access
2. **Access Control:** Use IP whitelisting, deploy VPN for remote access, implement fail2ban for rate limiting
3. **Monitoring & Detection:** Deploy SIEM for real-time alerting on failed login attempts, unusual authentication patterns, and port scanning activity
4. **System Hardening:** Keep SSH updated, configure strong ciphers, limit MaxAuthTries, and deploy host-based intrusion detection
5. **Deception Technology:** Expand honeypot deployment to detect reconnaissance early and integrate with threat intelligence platforms

Conclusion

This exercise validated that honeypots combined with traffic analysis provide comprehensive attack visibility. The integration of Nmap reconnaissance, Wireshark packet capture, and Cowrie behavioral logging demonstrated layered detection capabilities essential for modern security operations. Key takeaway: defense-in-depth with continuous monitoring enables rapid threat detection and response.

Supporting Evidence: nmap_scan_results.txt, cowrie_attack_capture.pcapng, cowrie_honeypot_screenshot.png, log excerpts

Classification: Lab Exercise | **Distribution:** Bincom Academy Instructors